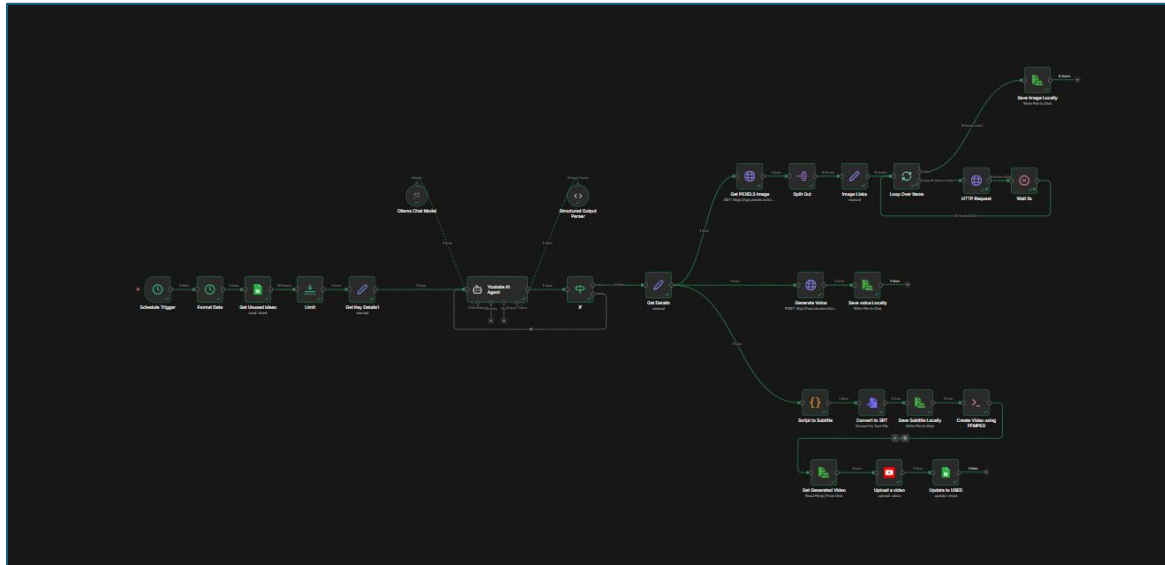


YOUTUBE AUTOMATION

USING n8n

Overview



This workflow automates YouTube content and uploading by using the scraped Ideas from reddit that is stored in a google sheet. Once idea is selected it generates a complete package including the script, title, description, keywords, and tags, streamlining the entire pre-production process for YouTube videos.

Problem

Brainstorming ideas, and script writing takes a lot of work and effort. It's also difficult when you hit a creative wall which can lead to delays in content production, inconsistent posting, and creative fatigue.

Solution

I built an automated workflow that helps streamline the entire content creation process. It generates content ideas, scripts, titles, keywords, tags, and descriptions automatically reducing the time needed for brainstorming and writing.

All generated ideas are stored in a Google Sheet for easy tracking. Each idea is marked with a status such as "NEW" or "USED," so nothing is repeated and content production stays organized and consistent.

Workflow Breakdown

1. The workflow will trigger at a specific time based on the configured schedule.
2. Generate scripts, titles, keywords, tags, and descriptions based on the idea retrieved from the Google Sheet.
3. Retrieve images from an API based on the generated keywords and save them locally.
4. Generate voiceovers based on the generated script and save them locally.
5. Generate subtitles based on the generated script and save them locally.
6. Generate the video using the images, voiceover, and subtitles, then save it locally.
7. Update the used title in the Google Sheet to "**USED**".

Tools & Integration

1. n8n
2. Google Sheet
3. Apify Reddit Scrape API
4. Kokoro TTS
5. FFMPEG
6. llama 3.2B AI Model
7. PEXELS API

Key Features

1. Real-time automation
2. Error Handling
3. Custom Schedule to generate contents

Results & Impact

1. Reduced time spent brainstorming ideas

From hours per week manually brainstorming, researching ideas, and taking notes to just seconds, since ideas are auto-scraped from Reddit and queued in the Google Sheet.

2. Consistent scheduled posting

Videos are generated and automatically marked "USED" on a fixed schedule, removing the gaps caused by missed days.

3. Saved hours of work

From an estimated 21 hours doing it manually; idea → script → voiceover → edit to almost 2 minutes of automated runtime.

4. Reduced creators' creative fatigue

Creators no longer have to generate scripts, source images, or edit manually; the workflow handles the repetitive tasks so creators can focus on strategy, review, or other channels.

Challenges

While building this workflow, I experienced a lot of issues, especially when trying to figure out where to get ideas that the entire package scripts, titles, keywords, descriptions, and tags could be based on. I searched for the perfect place to gather ideas and realized that Reddit has many communities where valuable ideas can come from. So, I tried integrating Reddit into my workflow and started with it.

But here's the problem: Reddit restricted its API, and the services I needed could no longer be used. So, I hit a wall again and realized that I could use Apify to scrape ideas from Reddit instead.

Another challenge I faced while doing this was finding the best text-to-speech model. I tried ElevenLabs, but it was quite expensive, so I looked for alternatives. After countless hours of searching, I found a text-to-speech solution that suited the workflow I was building.

When I had finally generated the scripts, tags, titles, keywords, audio files, images, and so on, I hit another wall: how can I combine all this media to generate a video? So, I again searched for a free service that I could use to stitch images and audio files into a single video. After hours of searching, I found FFmpeg, a multimedia framework used to record, convert, stream, and edit multimedia files. It is also a free, open-source tool that uses the command line, like PowerShell or CMD, which makes it perfect for an n8n Exec node.

After fixing all these issues with the tools I needed, I can now generate a video from just an idea from Reddit.

Learnings

Learned how to use specific TTS, and FFMPEG directly on Docker

Future Improvements

1. **Telegram Integration**

Allow users to send video ideas through Telegram and automatically create and post the video to YouTube.

2. **User Approval**

Let users review and approve the video before posting it to YouTube.

3. **Cloud Storage**

Store generated videos and media files in the cloud instead of locally.

4. **Format Selection**

Allow users to choose whether the generated video is a YouTube Short or a standard upload.

Conclusions

This project shows how automation can help creators work faster and focus on other things that matter to them. It can also help reduce the creative fatigue they experience.